

PRETORIA IRENE, South Africa

WMO: 682630

Lat: 25.9106S

Lon: 28.2106E

Elev: 5007

StdP: 12.22

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.0	39.2	10.0	11.1	59.4	13.4	13.1	58.4	20.4	56.2	18.6	57.4	7.5	180	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	88.4	61.0	86.2	61.0	84.2	61.3	68.3	78.8	67.4	77.4	66.5	76.2	9.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	113.7	71.4	64.5	109.8	70.9	63.6	106.4	70.5	35.9	78.8	35.0	77.5	34.3	76.1	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.3	14.3	DB	32.6	92.2	3.5	2.9	30.1	94.3	28.1	95.9	26.2	97.6	23.6	99.7	(4)
(5)			WB	27.4	70.4	3.1	2.2	25.1	71.9	23.3	73.2	21.5	74.4	19.3	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	70.8	70.7	68.6	63.8	59.1	54.7	53.8	58.8	65.0	68.0	68.8	70.3	(6)
(7)		DBStd	7.68	3.91	3.59	4.09	4.38	4.40	4.06	4.46	5.72	6.30	5.83	5.25	4.39	(7)
(8)		HDD50	38	0	0	0	0	3	9	18	7	1	0	0	0	(8)
(9)		HDD65	1286	6	3	14	72	186	308	349	202	78	38	23	7	(9)
(10)		CDD50	5267	646	581	576	414	284	150	134	280	450	558	563	631	(10)
(11)		CDD65	1040	187	164	125	36	2	0	0	11	76	130	136	173	(11)
(12)		CDH74	7641	1132	1000	804	268	46	6	3	145	784	1224	1062	1167	(12)
(13)		CDH80	1846	271	224	166	29	1	0	0	9	185	360	285	316	(13)
(14)	Wind	WSAvg	6.9	7.0	6.6	6.3	5.9	6.0	6.2	6.5	7.0	7.7	8.0	8.0	7.4	(14)
(15)	Precipitation	PrecAvg	27.2	4.9	4.1	3.8	1.5	0.6	0.3	0.1	0.2	0.6	2.4	3.8	5.0	(15)
(16)		PrecMax	44.2	9.1	12.2	12.9	4.6	3.9	1.3	0.5	1.2	2.8	6.6	7.8	7.5	(16)
(17)		PrecMin	17.6	1.8	0.9	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.7	1.7	(17)
(18)		PrecStd	6.6	2.0	2.6	2.7	1.1	0.9	0.4	0.1	0.3	0.7	1.6	1.7	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	88.9	87.2	83.3	78.2	74.0	73.8	81.0	87.8	90.7	89.8	90.9	(19)
(20)			MCWB	63.1	63.0	61.4	60.8	54.9	50.2	48.5	54.4	55.9	58.0	59.8	64.0	(20)
(21)		2%	DB	85.9	85.4	84.3	80.2	75.1	70.9	70.9	77.9	84.8	87.1	86.1	86.8	(21)
(22)			MCWB	63.9	63.8	62.5	59.2	53.8	49.7	48.3	53.0	55.0	57.9	60.9	63.7	(22)
(23)		5%	DB	83.0	82.6	81.4	77.3	73.1	68.4	68.3	75.0	82.1	84.1	83.4	83.4	(23)
(24)			MCWB	64.4	64.2	62.0	59.0	53.4	49.2	48.1	51.9	54.7	58.2	60.9	63.9	(24)
(25)		10%	DB	80.4	80.3	78.7	74.3	70.5	66.1	65.8	71.9	79.0	81.2	80.4	80.4	(25)
(26)			MCWB	64.8	64.2	62.1	58.2	52.8	48.5	47.7	50.7	54.2	57.9	61.1	63.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.8	69.1	68.3	65.1	61.1	55.1	53.6	58.6	60.7	64.3	67.3	69.2	(27)
(28)			MCD B	81.2	80.1	77.7	75.1	70.6	66.1	65.6	72.9	76.7	77.0	79.2	79.7	(28)
(29)		2%	WB	68.4	67.9	66.8	63.6	57.9	53.1	51.8	56.4	59.2	62.9	65.8	67.5	(29)
(30)			MCD B	78.6	78.2	75.5	73.4	68.3	64.8	63.9	71.9	75.1	76.4	77.4	78.8	(30)
(31)		5%	WB	67.4	66.9	65.6	62.1	56.2	51.6	50.2	54.6	57.8	61.8	64.6	66.5	(31)
(32)			MCD B	77.0	76.6	74.2	71.2	66.9	63.3	62.8	69.2	74.0	75.4	76.1	77.3	(32)
(33)		10%	WB	66.4	66.0	64.4	60.7	54.7	50.3	48.9	52.9	56.4	60.6	63.5	65.4	(33)
(34)			MCD B	75.5	75.1	73.1	69.3	65.3	62.1	62.2	66.9	72.5	74.1	74.3	76.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.7	18.3	18.4	19.5	21.9	23.2	23.3	23.9	23.8	22.3	19.7	18.4	(35)
(36)			MCDBR	22.1	22.1	22.7	23.1	24.3	25.5	25.7	26.0	26.1	25.4	23.7	22.5	(36)
(37)			MCWBR	7.7	7.5	7.8	8.5	9.8	11.1	10.9	10.9	9.7	8.6	7.8	7.3	(37)
(38)		5% WB	MCD BR	17.8	17.6	17.7	18.2	20.1	22.3	21.9	22.9	23.2	21.8	19.7	18.9	(38)
(39)			MCWBR	7.4	7.0	7.2	8.0	9.3	10.7	10.3	10.4	9.5	8.7	8.1	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.389	0.391	0.380	0.333	0.297	0.279	0.281	0.335	0.375	0.408	0.370	0.377	(40)	
(41)		taud	2.325	2.319	2.329	2.431	2.507	2.532	2.515	2.355	2.258	2.202	2.349	2.350	(41)	
(42)		Ebn at Noon	303	297	291	291	290	289	293	286	287	289	306	307	(42)	
(43)		Edh at Noon	43	43	40	33	28	26	27	35	43	48	42	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2103	1995	1767	1514	1388	1259	1367	1609	1913	2053	2135	2150	(44)	
(45)		RadStd	218	178	162	134	71	62	35	73	89	138	194	127	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (11 neighbors)		+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	-326	+421	+157	

Nomenclature: See separate page